

BENTLEY
AutoPIPE Advanced 12.06.00.48

AKABEL._ASME B31.3
08/04/2022
07:31 AM

BENTLEY
AutoPIPE Advanced 12.06.00.48

**
** AUTOPIPE SYSTEM INFORMATION **
**

SYSTEM NAME : AKABEL._ASME B31.3

PROJECT ID : AKABEL

PREPARED BY : _____
I.O

CHECKED BY : _____
E N

1ST APPROVER : _____

2ND APPROVER : _____

PIPING CODE : ASME B31.3

YEAR : 2016

VERTICAL AXIS : Y

AMBIENT TEMPERATURE : 24.0 deg C

COMPONENT LIBRARY : AUTOPIPE

MATERIAL LIBRARY : B313-16

MODEL REVISION NUMBER : 6

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COORDINATES DATA LISTING

POINT NAME	-----COORDINATE (mm) X	Y	Z	Mass Points
*** SEGMENT C, LINE # AKAB-A-100_N9				
B04	44851.20	7500.00	0.00	0
C01	44851.20	8200.00	0.00	0
*** SEGMENT D, LINE # AKABP-12"-P-014-C06				
C01	44851.20	8200.00	0.00	0
D01	44851.20	8220.00	0.00	0
D01 M	44851.20	8702.47	0.00	0
D02	44851.20	9184.95	0.00	0
D03 N	44851.20	9301.00	0.00	0
D03	44851.20	9758.20	0.00	0
D03 F	44851.20	9758.20	-457.20	0
D04	44851.20	9758.20	-4200.00	0
D05 N	44851.20	9758.20	-5142.80	0
D05	44851.20	9758.20	-5600.00	0
D05 F	44851.20	9301.00	-5600.00	0
D06 N	44851.20	7957.20	-5600.00	0
D06	44851.20	7500.00	-5600.00	0
D06 F	44394.00	7500.00	-5600.00	0
D11	43501.20	7500.00	-5600.00	0
D07	39631.20	7500.00	-5600.00	0
D08	37901.20	7500.00	-5600.00	0
D08 M	37418.73	7500.00	-5600.00	0
D09	36936.25	7500.00	-5600.00	0
D12	35636.25	7500.00	-5600.00	0
E03	32348.40	7500.00	-5600.00	0
*** SEGMENT E, LINE # AKABP-36"-P-013-C06				
O12	32348.40	7500.00	0.00	0
E01	32348.40	7500.00	-2800.00	0
E01 M	32348.40	7500.00	-3943.00	0
E02	32348.40	7500.00	-5086.00	0
E03	32348.40	7500.00	-5600.00	0
E04	32348.40	7500.00	-7460.00	0
*** SEGMENT G, LINE # AKAB-A-100_N4				
B01	39631.20	7500.00	0.00	0
G01	39631.20	8200.00	0.00	0
*** SEGMENT H, LINE # AKABP-4"-P-015-C06				
G01	39631.20	8200.00	0.00	0
H01	39631.20	8220.00	0.00	0
H01 M	39631.20	8448.47	0.00	0
H02	39631.20	8676.95	0.00	0
H03 N	39631.20	8824.55	0.00	0
H03	39631.20	8976.95	0.00	0
H03 F	39631.20	8976.95	-152.40	0
H06	39631.20	8976.95	-2800.00	0
H04 N	39631.20	8976.95	-5447.60	0
H04	39631.20	8976.95	-5600.00	0
H04 F	39631.20	8824.55	-5600.00	0
D07	39631.20	7500.00	-5600.00	0

C O O R D I N A T E S D A T A L I S T I N G

POINT NAME	-----COORDINATE (mm) -----			Mass Points
	X	Y	Z	

*** SEGMENT I, LINE # AKAB-A-100_N7

B05	47751.20	7500.00	0.00	0
I01	47751.20	8200.00	0.00	0

*** SEGMENT J, LINE # AKABP-4"-P-016-C06

I01	47751.20	8200.00	0.00	0
J01	47751.20	8220.00	0.00	0
J01 M	47751.20	8448.47	0.00	0
J02	47751.20	8676.95	0.00	0
J03 N	47751.20	8824.55	0.00	0
J03	47751.20	8976.95	0.00	0
J03 F	47751.20	8976.95	152.40	0
J16	47751.20	8976.95	3000.00	0
J04 N	47751.20	8976.95	3947.60	0
J04	47751.20	8976.95	4100.00	0
J04 F	47598.80	8976.95	4100.00	0
J11	44271.20	8976.95	4100.00	0
J12	40791.20	8976.95	4100.00	0
J13	37311.20	8976.95	4100.00	0
J14	33831.21	8976.95	4100.00	0
J15	31631.21	8976.95	4100.00	0
J05 N	30503.60	8976.95	4100.00	0
J05	30351.20	8976.95	4100.00	0
J05 F	30351.20	9129.35	4100.00	0
J06 N	30351.20	9474.55	4100.00	0
J06	30351.20	9626.95	4100.00	0
J06 F	30198.80	9626.95	4100.00	0
J09	29751.20	9626.95	4100.00	0
J10	29451.20	9626.95	4100.00	0
J07	29151.20	9626.95	4100.00	0
J07 M	29001.20	9826.95	4100.00	0
K09	28851.20	10026.95	4100.00	0

*** SEGMENT K, LINE # AKABP-6"-B-005-B06

K00	26056.09	9651.95	2557.00	0
K10	26056.09	9651.95	2807.00	0
K11	26056.09	9651.95	3453.50	0
K07	26056.09	9651.95	4100.00	0
K02 N	26056.09	9798.35	4100.00	0
K02	26056.09	10026.95	4100.00	0
K02 F	26284.69	10026.95	4100.00	0
K03	26406.09	10026.95	4100.00	0
K12	26956.09	10026.95	4100.00	0
K04	27506.09	10026.95	4100.00	0
K04 M	27734.69	10026.95	4100.00	0
K05	27963.29	10026.95	4100.00	0
K06	28716.20	10026.95	4100.00	0
K08	28821.20	10026.95	4100.00	0
K09	28851.20	10026.95	4100.00	0

*** SEGMENT L, LINE # AKABP-4"-P-016-C06

C O O R D I N A T E S D A T A L I S T I N G

POINT NAME	-----COORDINATE (mm) -----			Mass Points
	X	Y	Z	
J09	29751.20	9626.95	4100.00	0
L01 N	29751.20	9626.95	3852.40	0
L01	29751.20	9626.95	3700.00	0
L01 F	29598.80	9626.95	3700.00	0
L08	29076.20	9626.95	3700.00	0
L02	28401.20	9626.95	3700.00	0
L02 M	28172.73	9626.95	3700.00	0
L03	27944.25	9626.95	3700.00	0
L03 M	27715.65	9626.95	3700.00	0
L04	27487.05	9626.95	3700.00	0
L09	26946.57	9626.95	3700.00	0
L05 N	26558.48	9626.95	3700.00	0
L05	26406.08	9626.95	3700.00	0
L05 F	26406.08	9779.35	3700.00	0
L06 N	26406.08	9874.55	3700.00	0
L06	26406.08	10026.95	3700.00	0
L06 F	26406.08	10026.95	3852.40	0
K03	26406.09	10026.95	4100.00	0

*** SEGMENT M, LINE # AKABP-6"-B-001-B06

K07	26056.09	9651.95	4100.00	0
M09	26056.09	9651.95	4800.00	0
M08	26056.09	9651.95	6750.00	0
M01 N	26056.09	9651.95	7021.40	0
M01	26056.09	9651.95	7250.00	0
M01 F	25877.58	9509.14	7250.00	0
M02 N	25368.70	9102.04	7250.00	0
M02	25306.09	9051.95	7250.00	0
M02 F	25225.90	9051.95	7250.00	0
M07	23256.09	9051.95	7250.00	0
M06	18256.09	9051.95	7250.00	0
M05	13256.09	9051.95	7250.00	0
M04	8256.09	9051.95	7250.00	0
M03	3256.09	9051.95	7250.00	0

*** SEGMENT N, LINE # AKABP-6"-B-001-B06

O11	29848.40	7500.00	0.00	0
N04	29848.40	7900.00	0.00	0
N01	29848.40	8000.00	0.00	0
N01 M	29848.40	8184.02	0.00	0
N02	29848.40	8368.05	0.00	0
N03 N	29848.40	8741.85	0.00	0
N03	29848.40	8818.04	0.00	0
N03 F	29848.40	8818.04	76.20	0
N16	29848.40	8818.04	2000.00	0
N05 N	29848.40	8818.04	2730.80	0
N05	29848.40	8818.04	2807.00	0
N05 F	29848.40	8894.25	2807.00	0
N11 N	29848.40	9575.75	2807.00	0
N11	29848.40	9651.95	2807.00	0
N11 F	29772.20	9651.95	2807.00	0
N06	29498.40	9651.95	2807.00	0

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C O O R D I N A T E S D A T A L I S T I N G

POINT	-----COORDINATE (mm)-----			Mass
NAME	X	Y	Z	Points
N06 M	29314.25	9651.95	2807.00	0
N07	29130.10	9651.95	2807.00	0
N15	28655.10	9651.95	2807.00	0
N08	28180.10	9651.95	2807.00	0
N08 M	27996.08	9651.95	2807.00	0
N09	27812.05	9651.95	2807.00	0
N10	27412.05	9651.95	2807.00	0
N12	27142.05	9651.95	2807.00	0
N14	26624.07	9651.95	2807.00	0
K10	26056.09	9651.95	2807.00	0

PIPE DATA LISTING

Pipe ID/ Material CladMaterial ---Line Class---	Nom/ Sch	O.D. mm	-----Thickness(mm)-----				Spec	InsuDen/ LingDen/ CladDen kg/m3	Weight(N/m) Pipe/ Ling/ Insu/ Clad	E/ W/ f	Composition/ Steel Group/ Tcr			
			W.Th. Clad	Corr	Mill	Insu	Ling	Grav/ InsMt		Total				
Tag No. : <None> 12"C06 API5L-B 900	300 140	323.85	28.57	6.00	3.57	0	0	0.70	0.000 0.000 0.000	2036 383 0	2419	1.00 1.00 1.00	Carbon Carbon 0.000	Steel
Tag No. : <None> 36"C06 API5L-X60 900	900 NS	914.40	44.43	6.00	5.55	0	0	0.70	0.000 0.000 0.000	9328 3673 0	13001	1.00 1.00 1.00	Carbon Carbon 0.000	Steel
Tag No. : <None> 4"C06 API5L-B 900	100 XXS	114.30	17.12	6.00	2.14	0	0	0.70	0.000 0.000 0.000	401 34.54 0	436	1.00 1.00 1.00	Carbon Carbon 0.000	Steel
Tag No. : <None> 6"B06 API5L-B 150	150 XS	168.27	10.97	6.00	1.37	0	0	0.70	0.000 0.000 0.000	417 115 0	532	1.00 1.00 1.00	Carbon Carbon 0.000	Steel
Tag No. : <None> 4"B06 API5L-B 150	100 120	114.30	11.13	6.00	1.39	0	0	0.70	0.000 0.000 0.000	277 45.66 0	323	1.00 1.00 1.00	Carbon Carbon 0.000	Steel
Tag No. : <None> 2"C06 API5L-B 900	50 XXS	60.325	11.07	6.00	1.38	0	0	0.70	0.000 0.000 0.000	132 7.85 0	139	1.00 1.00 1.00	Carbon Carbon 0.000	Steel
Tag No. : <None> 2"B06 API5L-B 150	50 XXS	60.325	11.07	6.00	1.38	0	0	0.70	0.000 0.000 0.000	132 7.85 0	139	1.00 1.00 1.00	Carbon Carbon 0.000	Steel

M A T E R I A L D A T A L I S T I N G

Material Name	Pipe ID	Density kg/m3	Pois. Ratio	Temper. deg C	Modulus E6 N/mm2 Axial	Modulus E6 N/mm2 Hoop	Modulus E6 N/mm2 Shear	Expans. mm/m	Composition
API5L-B	12"C06	7833.0	0.30	24.0 100.0 -29.0 23.0	0.20320 0.19816 0.20593 0.20327	0.20320	0.07815	0.8900 -0.5971 -0.0115	Carbon Steel
API5L-X60	36"C06	7833.0	0.30	24.0 60.0 12.7 23.0 100.0 -29.0	0.20320 0.20080 0.20382 0.20327 0.19816 0.20593	0.20320	0.07815	0.4154 -0.1280 -0.0115 0.8900 -0.5971	Carbon Steel
API5L-B	4"C06	7833.0	0.30	24.0 100.0 -29.0 23.0 60.0	0.20320 0.19816 0.20593 0.20327 0.20080	0.20320	0.07815	0.8900 -0.5971 -0.0115 0.4154	Carbon Steel
API5L-B	6"B06	7833.0	0.30	24.0 60.0 -29.0 23.0	0.20320 0.20080 0.20593 0.20327	0.20320	0.07815	0.4154 -0.5971 -0.0115	Carbon Steel
API5L-B	4"B06	7833.0	0.30	24.0 60.0 -29.0 23.0	0.20320 0.20080 0.20593 0.20327	0.20320	0.07815	0.4154 -0.5971 -0.0115	Carbon Steel
API5L-B	2"C06	7833.0	0.30	24.0 100.0 -29.0 23.0	0.20320 0.19816 0.20593 0.20327	0.20320	0.07815	0.8900 -0.5971 -0.0115	Carbon Steel
API5L-B	2"B06	7833.0	0.30	24.0 60.0 -29.0 23.0	0.20320 0.20080 0.20593 0.20327	0.20320	0.07815	0.4154 -0.5971 -0.0115	Carbon Steel

M A T E R I A L A L L O W A B L E D A T A L I S T I N G

Material Name	Pipe ID	Temper. deg C	Allow. N/mm2	Yield N/mm2	Ultimate Weld Reduction Factor W
API5L-B	12"C06	24.0	137.9	241.3	413.7
		100.0	137.9	220.4	1.000
		-29.0	137.9	241.3	1.000
		23.0	137.9	241.3	1.000
API5L-X60	36"C06	24.0	172.4	413.7	517.1
		60.0	172.4	0.0	1.000
		12.7	172.4	0.0	1.000
		23.0	172.4	0.0	1.000
		100.0	172.4	0.0	1.000
API5L-B	4"C06	-29.0	172.4	0.0	1.000
		24.0	137.9	241.3	413.7
		100.0	137.9	220.4	1.000
		-29.0	137.9	241.3	1.000
		23.0	137.9	241.3	1.000
API5L-B	6"B06	60.0	137.9	233.3	1.000
		24.0	137.9	241.3	413.7
		60.0	137.9	233.3	1.000
		-29.0	137.9	241.3	1.000
		23.0	137.9	241.3	1.000
API5L-B	4"B06	24.0	137.9	241.3	413.7
		60.0	137.9	233.3	1.000
		-29.0	137.9	241.3	1.000
		23.0	137.9	241.3	1.000
		24.0	137.9	241.3	413.7
API5L-B	2"C06	100.0	137.9	220.4	1.000
		-29.0	137.9	241.3	1.000
		23.0	137.9	241.3	1.000
		24.0	137.9	241.3	413.7
API5L-B	2"B06	60.0	137.9	233.3	1.000
		-29.0	137.9	241.3	1.000
		23.0	137.9	241.3	1.000
		24.0	137.9	241.3	413.7

OPERATING TEMPERATURE AND PRESSURE DATA
STRESSES IN N/mm2

POINT NAME	CASE	PRESS. N/mm2	TEMPER deg C	EXPAN. mm/m	MODULUS E6 N/mm2	ALLOW STRESS	YIELD STRESS
***	SEGMENT C, LINE # AKAB-A-100 N9						
B04	T1	11.0000	100	0.890	0.19816	137.90	220.41
	T2	11.0000	-29.00	-0.597	0.20593	137.90	241.32
	T3	11.0000	23.00	-0.012	0.20327	137.90	241.32
C01	Same as previous point.						
***	SEGMENT D, LINE # AKABP-12"-P-014-C06						
C01	T1	11.0000	100	0.890	0.19816	137.90	220.41
	T2	11.0000	-29.00	-0.597	0.20593	137.90	241.32
	T3	11.0000	23.00	-0.012	0.20327	137.90	241.32
E03	Same as previous point.						
***	SEGMENT E, LINE # AKABP-36"-P-013-C06						
O12	T1	11.0000	60.00	0.415	0.20080	172.37	0
	T2	11.0000	12.70	-0.128	0.20382	172.37	0
	T3	11.0000	23.00	-0.012	0.20327	172.37	0
E01	T1	11.0000	100	0.890	0.19816	172.37	0
	T2	11.0000	-29.00	-0.597	0.20593	172.37	0
	T3	11.0000	23.00	-0.012	0.20327	172.37	0
E04	Same as previous point.						
***	SEGMENT G, LINE # AKAB-A-100 N4						
B01	T1	11.0000	100	0.890	0.19816	137.90	220.41
	T2	11.0000	-29.00	-0.597	0.20593	137.90	241.32
	T3	11.0000	23.00	-0.012	0.20327	137.90	241.32
G01	Same as previous point.						
***	SEGMENT H, LINE # AKABP-4"-P-015-C06						
G01	T1	11.0000	100	0.890	0.19816	137.90	220.41
	T2	11.0000	-29.00	-0.597	0.20593	137.90	241.32
	T3	11.0000	23.00	-0.012	0.20327	137.90	241.32
D07	Same as previous point.						
***	SEGMENT I, LINE # AKAB-A-100 N7						
B05	T1	11.0000	100	0.890	0.19816	137.90	220.41
	T2	11.0000	-29.00	-0.597	0.20593	137.90	241.32
	T3	11.0000	23.00	-0.012	0.20327	137.90	241.32
I01	Same as previous point.						
***	SEGMENT J, LINE # AKABP-4"-P-016-C06						
I01	T1	11.0000	100	0.890	0.19816	137.90	220.41
	T2	11.0000	-29.00	-0.597	0.20593	137.90	241.32
	T3	11.0000	23.00	-0.012	0.20327	137.90	241.32
K09	Same as previous point.						
***	SEGMENT K, LINE # AKABP-6"-B-005-B06						
K00	T1	1.9600	60.00	0.415	0.20080	137.90	233.32
	T2	1.9600	-29.00	-0.597	0.20593	137.90	241.32
	T3	1.9600	23.00	-0.012	0.20327	137.90	241.32
K09	Same as previous point.						
***	SEGMENT L, LINE # AKABP-4"-P-016-C06						
J09	T1	11.0000	100	0.890	0.19816	137.90	220.41
	T2	11.0000	-29.00	-0.597	0.20593	137.90	241.32
	T3	11.0000	23.00	-0.012	0.20327	137.90	241.32
I04	T1	1.9600	60.00	0.415	0.20080	137.90	233.32
	T2	1.9600	-29.00	-0.597	0.20593	137.90	241.32
	T3	1.9600	23.00	-0.012	0.20327	137.90	241.32

OPERATING TEMPERATURE AND PRESSURE DATA
STRESSES IN N/mm2

POINT NAME	CASE	PRESS. N/mm2	TEMPER deg C	EXPAN. mm/m	MODULUS E6 N/mm2	ALLOW STRESS	YIELD STRESS
K03	Same as previous point.						
*** SEGMENT M, LINE # AKABP-6"-B-001-B06							
K07	T1	1.9600	60.00	0.415	0.20080	137.90	233.32
	T2	1.9600	-29.00	-0.597	0.20593	137.90	241.32
	T3	1.9600	23.00	-0.012	0.20327	137.90	241.32
M03	Same as previous point.						
*** SEGMENT N, LINE # AKABP-6"-B-001-B06							
O11	T1	11.0000	100	0.890	0.19816	137.90	220.41
	T2	11.0000	-29.00	-0.597	0.20593	137.90	241.32
	T3	11.0000	23.00	-0.012	0.20327	137.90	241.32
N09	T1	1.9600	60.00	0.415	0.20080	137.90	233.32
	T2	1.9600	-29.00	-0.597	0.20593	137.90	241.32
	T3	1.9600	23.00	-0.012	0.20327	137.90	241.32
K10	Same as previous point.						

u User-defined value
* Non-code material for allowable stress;
Non-standard material for expansion and modulus

S E G M E N T D A T A L I S T I N G

Segment Name	First Node	Last Node	Line Number	Apply Wind	Apply Bowling	Apply Buoyancy
C	B04	C01	AKAB-A-100 N9	Yes	No	No
D	C01	E03	AKABP-12"-P-014-C06	Yes	No	No
E	O12	E04	AKABP-36"-P-013-C06	Yes	No	No
G	B01	G01	AKAB-A-100 N4	Yes	No	No
H	G01	D07	AKABP-4"-P-015-C06	Yes	No	No
I	B05	I01	AKAB-A-100 N7	Yes	No	No
J	I01	K09	AKABP-4"-P-016-C06	Yes	No	No
K	K00	K09	AKABP-6"-B-005-B06	Yes	No	No
L	J09	K03	AKABP-4"-P-016-C06	Yes	No	No
M	K07	M03	AKABP-6"-B-001-B06	Yes	No	No
N	O11	K10	AKABP-6"-B-001-B06	Yes	No	No

L O A D S S U M M A R Y D A T A L I S T I N G

WIND LOAD CASES :

Number of load cases : 4

Ground elevation for wind : 2000.00 mm

Wind shape factor multiplier for piping: 1.000

Wind shape factor multiplier for beams: 1.000

Exposure factor for buried pipes : 0.000

Exposure factor for submerged pipes : 0.000

Exposed Segments - C D E G H I J K L M N

Wind application method : Projected

L O A D S S U M M A R Y D A T A L I S T I N G

Load case 1 - W1

Direction - Global X-axis

ASCE 7-2010

Basic wind speed at 33 ft (10m): 136.79 kmph

Exposure category : B

Gust effect factor (G) : 0.85

Force coefficient (Cf), Figure 29.5-1 : Automatic

ASCE 7-10 PARAMETERS AS A FUNCTION OF HEIGHT (PIPING)

HEIGHT (mm)	Kz	Cf	Qz (N/m2)	Pressure (N/m2)
0.6	0.575	0.700	508.965	302.834
3048.0	0.575	0.700	508.965	302.834
6096.0	0.624	0.700	552.567	328.777
10058.4	0.720	0.700	637.563	379.350
15240.0	0.811	0.700	717.930	427.168
22860.0	0.910	0.700	806.109	479.635
30480.0	0.988	0.700	875.166	520.724
60960.0	1.205	0.700	1066.839	634.770
152400.0	1.565	0.700	1386.105	824.733
304800.0	1.908	0.700	1689.681	1005.360

Kz = Velocity pressure exposure coefficient

Cf = Force coefficient or wind shape factor

Qz = Velocity pressure = $0.00256 \cdot Kz \cdot Kzt \cdot V^2$ (Eq. 27.3-1)

Pressure = $Qz \cdot G \cdot Cf$ (Eq. 23)

Per ASCE Figure 29.5-1, wind appl. method should be "Projected"

Default Gust Factor G assumes pipe frequency > 1 Hz.

Piping is assumed not to be on an isolated hill (Kzt=1).

Automatic Cf reported in chart above is calculated

assuming D=1 ft. Program uses actual Pipe OD to

calculate Cf at height during analysis.

L O A D S S U M M A R Y D A T A L I S T I N G

Load case 1 - W1

Structural members wind load application data

Structural members wind facing width: Wind facing

Gust effect factor (G) for structures (manual): 0.85

Force coefficient (Cf) for structures (manual): 2.000

ASCE 7-10 PARAMETERS AS A FUNCTION OF HEIGHT (STRUCTURE)

HEIGHT (mm)	Kz	Cf	Qz (N/m2)	Pressure (N/m2)
0.6	0.575	2.000	508.965	865.241
3048.0	0.575	2.000	508.965	865.241
6096.0	0.624	2.000	552.567	939.364
10058.4	0.720	2.000	637.563	1083.857
15240.0	0.811	2.000	717.930	1220.481
22860.0	0.910	2.000	806.109	1370.385
30480.0	0.988	2.000	875.166	1487.782
60960.0	1.205	2.000	1066.839	1813.627
152400.0	1.565	2.000	1386.105	2356.379
304800.0	1.908	2.000	1689.681	2872.458

Kz = Velocity pressure exposure coefficient

Cf = Force coefficient or wind shape factor

Qz = Velocity pressure = $0.00256 \cdot Kz \cdot Kzt \cdot V^2$ (Eq. 27.3-1)

Pressure = $Qz \cdot G \cdot Cf$ (Eq. 23)

L O A D S S U M M A R Y D A T A L I S T I N G

Load case 2 - W2

Direction - Global -X-axis

ASCE 7-2010

Basic wind speed at 33 ft (10m): 136.79 kmph

Exposure category : B

Gust effect factor (G) : 0.85

Force coefficient (Cf), Figure 29.5-1 : Automatic

ASCE 7-10 PARAMETERS AS A FUNCTION OF HEIGHT (PIPING)

HEIGHT (mm)	Kz	Cf	Qz (N/m2)	Pressure (N/m2)
0.6	0.575	0.700	508.965	302.834
3048.0	0.575	0.700	508.965	302.834
6096.0	0.624	0.700	552.567	328.777
10058.4	0.720	0.700	637.563	379.350
15240.0	0.811	0.700	717.930	427.168
22860.0	0.910	0.700	806.109	479.635
30480.0	0.988	0.700	875.166	520.724
60960.0	1.205	0.700	1066.839	634.770
152400.0	1.565	0.700	1386.105	824.733
304800.0	1.908	0.700	1689.681	1005.360

Kz = Velocity pressure exposure coefficient

Cf = Force coefficient or wind shape factor

Qz = Velocity pressure = $0.00256 \cdot Kz \cdot Kzt \cdot V^2$ (Eq. 27.3-1)

Pressure = $Qz \cdot G \cdot Cf$ (Eq. 23)

Per ASCE Figure 29.5-1, wind appl. method should be "Projected"

Default Gust Factor G assumes pipe frequency > 1 Hz.

Piping is assumed not to be on an isolated hill (Kzt=1).

Automatic Cf reported in chart above is calculated

assuming D=1 ft. Program uses actual Pipe OD to

calculate Cf at height during analysis.

L O A D S S U M M A R Y D A T A L I S T I N G

Load case 2 - W2

Structural members wind load application data

Structural members wind facing width: Wind facing

Gust effect factor (G) for structures (manual): 0.85

Force coefficient (Cf) for structures (manual): 2.000

ASCE 7-10 PARAMETERS AS A FUNCTION OF HEIGHT (STRUCTURE)

HEIGHT (mm)	Kz	Cf	Qz (N/m2)	Pressure (N/m2)
0.6	0.575	2.000	508.965	865.241
3048.0	0.575	2.000	508.965	865.241
6096.0	0.624	2.000	552.567	939.364
10058.4	0.720	2.000	637.563	1083.857
15240.0	0.811	2.000	717.930	1220.481
22860.0	0.910	2.000	806.109	1370.385
30480.0	0.988	2.000	875.166	1487.782
60960.0	1.205	2.000	1066.839	1813.627
152400.0	1.565	2.000	1386.105	2356.379
304800.0	1.908	2.000	1689.681	2872.458

Kz = Velocity pressure exposure coefficient

Cf = Force coefficient or wind shape factor

Qz = Velocity pressure = $0.00256 \cdot Kz \cdot Kzt \cdot V^2$ (Eq. 27.3-1)

Pressure = $Qz \cdot G \cdot Cf$ (Eq. 23)

L O A D S S U M M A R Y D A T A L I S T I N G

Load case 3 - W3

Direction - Global Z-axis

ASCE 7-2010

Basic wind speed at 33 ft (10m): 136.79 kmph

Exposure category : B

Gust effect factor (G) : 0.85

Force coefficient (Cf), Figure 29.5-1 : Automatic

ASCE 7-10 PARAMETERS AS A FUNCTION OF HEIGHT (PIPING)

HEIGHT (mm)	Kz	Cf	Qz (N/m2)	Pressure (N/m2)
0.6	0.575	0.700	508.965	302.834
3048.0	0.575	0.700	508.965	302.834
6096.0	0.624	0.700	552.567	328.777
10058.4	0.720	0.700	637.563	379.350
15240.0	0.811	0.700	717.930	427.168
22860.0	0.910	0.700	806.109	479.635
30480.0	0.988	0.700	875.166	520.724
60960.0	1.205	0.700	1066.839	634.770
152400.0	1.565	0.700	1386.105	824.733
304800.0	1.908	0.700	1689.681	1005.360

Kz = Velocity pressure exposure coefficient

Cf = Force coefficient or wind shape factor

Qz = Velocity pressure = $0.00256 \cdot Kz \cdot Kzt \cdot V^2$ (Eq. 27.3-1)

Pressure = $Qz \cdot G \cdot Cf$ (Eq. 23)

Per ASCE Figure 29.5-1, wind appl. method should be "Projected"

Default Gust Factor G assumes pipe frequency > 1 Hz.

Piping is assumed not to be on an isolated hill (Kzt=1).

Automatic Cf reported in chart above is calculated

assuming D=1 ft. Program uses actual Pipe OD to

calculate Cf at height during analysis.

L O A D S S U M M A R Y D A T A L I S T I N G

Load case 3 - W3

Structural members wind load application data

Structural members wind facing width: Wind facing

Gust effect factor (G) for structures (manual): 0.85

Force coefficient (Cf) for structures (manual): 2.000

ASCE 7-10 PARAMETERS AS A FUNCTION OF HEIGHT (STRUCTURE)

HEIGHT (mm)	Kz	Cf	Qz (N/m2)	Pressure (N/m2)
0.6	0.575	2.000	508.965	865.241
3048.0	0.575	2.000	508.965	865.241
6096.0	0.624	2.000	552.567	939.364
10058.4	0.720	2.000	637.563	1083.857
15240.0	0.811	2.000	717.930	1220.481
22860.0	0.910	2.000	806.109	1370.385
30480.0	0.988	2.000	875.166	1487.782
60960.0	1.205	2.000	1066.839	1813.627
152400.0	1.565	2.000	1386.105	2356.379
304800.0	1.908	2.000	1689.681	2872.458

Kz = Velocity pressure exposure coefficient

Cf = Force coefficient or wind shape factor

Qz = Velocity pressure = $0.00256 \cdot Kz \cdot Kzt \cdot V^2$ (Eq. 27.3-1)

Pressure = $Qz \cdot G \cdot Cf$ (Eq. 23)

L O A D S S U M M A R Y D A T A L I S T I N G

Load case 4 - W4

Direction - Global -Z-axis

ASCE 7-2010

Basic wind speed at 33 ft (10m): 136.79 kmph

Exposure category : B

Gust effect factor (G) : 0.85

Force coefficient (Cf), Figure 29.5-1 : Automatic

ASCE 7-10 PARAMETERS AS A FUNCTION OF HEIGHT (PIPING)

HEIGHT (mm)	Kz	Cf	Qz (N/m2)	Pressure (N/m2)
0.6	0.575	0.700	508.965	302.834
3048.0	0.575	0.700	508.965	302.834
6096.0	0.624	0.700	552.567	328.777
10058.4	0.720	0.700	637.563	379.350
15240.0	0.811	0.700	717.930	427.168
22860.0	0.910	0.700	806.109	479.635
30480.0	0.988	0.700	875.166	520.724
60960.0	1.205	0.700	1066.839	634.770
152400.0	1.565	0.700	1386.105	824.733
304800.0	1.908	0.700	1689.681	1005.360

Kz = Velocity pressure exposure coefficient

Cf = Force coefficient or wind shape factor

Qz = Velocity pressure = $0.00256 \cdot Kz \cdot Kzt \cdot V^2$ (Eq. 27.3-1)

Pressure = $Qz \cdot G \cdot Cf$ (Eq. 23)

Per ASCE Figure 29.5-1, wind appl. method should be "Projected"

Default Gust Factor G assumes pipe frequency > 1 Hz.

Piping is assumed not to be on an isolated hill (Kzt=1).

Automatic Cf reported in chart above is calculated

assuming D=1 ft. Program uses actual Pipe OD to

calculate Cf at height during analysis.

L O A D S S U M M A R Y D A T A L I S T I N G

Load case 4 - W4

Structural members wind load application data

Structural members wind facing width: Wind facing

Gust effect factor (G) for structures (manual): 0.85

Force coefficient (Cf) for structures (manual): 2.000

ASCE 7-10 PARAMETERS AS A FUNCTION OF HEIGHT (STRUCTURE)

HEIGHT (mm)	Kz	Cf	Qz (N/m2)	Pressure (N/m2)
0.6	0.575	2.000	508.965	865.241
3048.0	0.575	2.000	508.965	865.241
6096.0	0.624	2.000	552.567	939.364
10058.4	0.720	2.000	637.563	1083.857
15240.0	0.811	2.000	717.930	1220.481
22860.0	0.910	2.000	806.109	1370.385
30480.0	0.988	2.000	875.166	1487.782
60960.0	1.205	2.000	1066.839	1813.627
152400.0	1.565	2.000	1386.105	2356.379
304800.0	1.908	2.000	1689.681	2872.458

Kz = Velocity pressure exposure coefficient

Cf = Force coefficient or wind shape factor

Qz = Velocity pressure = $0.00256 \cdot Kz \cdot Kzt \cdot V^2$ (Eq. 27.3-1)

Pressure = $Qz \cdot G \cdot Cf$ (Eq. 23)

Point Name	Support Type	Support ID	Conn.to /Dir	Comp.Wt (kg)	Stiff.	Gap 1 (mm)	Gap 2 (mm)	Fric. Fact.	GapSet /#hgr	Preload (N)	Ld.Var	Size	Figure
---------------	-----------------	---------------	-----------------	-----------------	--------	---------------	---------------	----------------	-----------------	-----------------	--------	------	--------

Tag	No.:						Attachment	ID:			
D04	Guide-V	D04	1	Ground	0.000	Rigid	0.0	3.0	0.10	Weightless	
	Guide-H	D04	1	Ground	0.000	Rigid	3.0	3.0	0.10	Weightless	
Tag	No.:						Attachment	ID:			
D11	Shoe	D11	1	Ground	0.000	Rigid	0.0	2540.0	0.10	Weightless	
Tag	No.:						Attachment	ID:			
D12	Guide-V	D12	1	Ground	0.000	Rigid	0.0	3.0	0.10	Weightless	
	Guide-H	D12	1	Ground	0.000	Rigid	3.0	3.0	0.10	Weightless	
Tag	No.:						Attachment	ID:			
E01	M Shoe	E01	M1	Ground	0.000	Rigid	0.0	2540.0	0.00	Weightless	
Tag	No.:						Attachment	ID:			
H06	Guide-V	H06	1	Ground	0.000	Rigid	0.0	3.0	0.10	Weightless	
	Guide-H	H06	1	Ground	0.000	Rigid	3.0	3.0	0.10	Weightless	
Tag	No.:						Attachment	ID:			
J16	Guide-V	J16	1	Ground	0.000	Rigid	0.0	3.0	0.10	Weightless	
	Guide-H	J16	1	Ground	0.000	Rigid	3.0	3.0	0.10	Weightless	
Tag	No.:						Attachment	ID:			
J11	Shoe	J11	1	Ground	0.000	Rigid	0.0	2540.0	0.10	Weightless	
Tag	No.:						Attachment	ID:			
J12	Guide-V	J12	1	Ground	0.000	Rigid	0.0	3.0	0.10	Weightless	
	Guide-H	J12	1	Ground	0.000	Rigid	3.0	3.0	0.10	Weightless	
Tag	No.:						Attachment	ID:			
J13	Shoe	J13	1	Ground	0.000	Rigid	0.0	2540.0	0.10	Weightless	
Tag	No.:						Attachment	ID:			
J14	Guide-V	J14	1	Ground	0.000	Rigid	0.0	3.0	0.10	Weightless	
	Guide-H	J14	1	Ground	0.000	Rigid	3.0	3.0	0.10	Weightless	
Tag	No.:						Attachment	ID:			
J15	Guide-V	J15	1	Ground	0.000	Rigid	0.0	3.0	0.10	Weightless	
	Guide-H	J15	1	Ground	0.000	Rigid	3.0	3.0	0.10	Weightless	
Tag	No.:						Attachment	ID:			
K12	Shoe	K12	1	Ground	0.000	Rigid	0.0	2540.0	0.10	Weightless	
Tag	No.:						Attachment	ID:			
L08	Shoe	L08	1	Ground	0.000	Rigid	0.0	2540.0	0.10	Weightless	
Tag	No.:						Attachment	ID:			
L09	Guide-V	L09	1	Ground	0.000	Rigid	0.0	3.0	0.10	Weightless	
	Guide-H	L09	1	Ground	0.000	Rigid	3.0	3.0	0.10	Weightless	
Tag	No.:						Attachment	ID:			

S U P P O R T D A T A L I S T I N G

Point Name	Support Type	Support ID	Conn.to /Dir	Comp.Wt (kg)	Stiff.	Gap 1 (mm)	Gap 2 (mm)	Fric. Fact.	GapSet /#hgr	Preload (N)	Ld.Var	Size	Figure
M09	Guide-V	M09	1	Ground	0.000 Rigid	0.0	3.0	0.10	Weightless				
	Guide-H	M09	1	Ground	0.000 Rigid	3.0	3.0	0.10	Weightless				
Tag No.: <None>													
M08	Shoe	M08	1	Ground	0.000 Rigid	0.0	2540.0	0.10	Weightless				
Tag No.: <None>													
M07	Guide-V	M07	1	Ground	0.000 Rigid	0.0	3.0	0.10	Weightless				
	Guide-H	M07	1	Ground	0.000 Rigid	3.0	3.0	0.10	Weightless				
Tag No.: <None>													
M06	Guide-V	M06	1	Ground	0.000 Rigid	0.0	3.0	0.10	Weightless				
	Guide-H	M06	1	Ground	0.000 Rigid	3.0	3.0	0.10	Weightless				
Tag No.: <None>													
M05	Guide-V	M05	1	Ground	0.000 Rigid	0.0	3.0	0.10	Weightless				
	Guide-H	M05	1	Ground	0.000 Rigid	3.0	3.0	0.10	Weightless				
Tag No.: <None>													
M04	Guide-V	M04	1	Ground	0.000 Rigid	0.0	3.0	0.10	Weightless				
	Guide-H	M04	1	Ground	0.000 Rigid	3.0	3.0	0.10	Weightless				
Tag No.: <None>													
N16	Guide-V	N16	1	Ground	0.000 Rigid	0.0	3.0	0.10	Weightless				
	Guide-H	N16	1	Ground	0.000 Rigid	3.0	3.0	0.10	Weightless				
Tag No.: <None>													
N15	Guide-V	N15	1	Ground	0.000 Rigid	0.0	3.0	0.10	Weightless				
	Guide-H	N15	1	Ground	0.000 Rigid	3.0	3.0	0.10	Weightless				
Tag No.: <None>													
N14	Guide-V	N14	1	Ground	0.000 Rigid	0.0	3.0	0.10	Weightless				
	Guide-H	N14	1	Ground	0.000 Rigid	3.0	3.0	0.10	Weightless				

Spring Manufacturer: Anvil/Grinnell

NOTE 1: No soil supports present in the system.

Gap 1 : V-stop,Guide-V=down, Linestop,Incline,Tie/link=backward, Guide-H=Left

Gap 2 : V-stop,Guide-V=Up , Linestop,Incline,Tie/link=forward , Guide-H=Right

Stiffness units for rotation support: N.m/deg , all others: N/mm

B E N D D A T A L I S T I N G

Point Name	Bend Type	Radius (mm)	Angle (deg)	OD (mm)	tnom (mm)	Material	Flan^ /Cuts	SIF in	SIF out	Analysis Set	Flexibility (K)	Pressure Case
D03	Elbow	457L	90.0	323.9	28.6	API5L-B	1/0	1.16	1.00	1	2.53	None
								1.16	1.00	2	2.53	None
D05	Elbow	457L	90.0	323.9	28.6	API5L-B	0/0	1.27	1.06	1	2.75	None
								1.27	1.06	2	2.75	None
D06	Elbow	457L	90.0	323.9	28.6	API5L-B	0/0	1.27	1.06	1	2.75	None
								1.27	1.06	2	2.75	None
H03	Elbow	152L	90.0	114.3	17.1	API5L-B	0/0	1.00	1.00	1	1.49	None
								1.00	1.00	2	1.49	None
H04	Elbow	152L	90.0	114.3	17.1	API5L-B	0/0	1.00	1.00	1	1.49	None
								1.00	1.00	2	1.49	None
J03	Elbow	152L	90.0	114.3	17.1	API5L-B	0/0	1.00	1.00	1	1.49	None
								1.00	1.00	2	1.49	None
J04	Elbow	152L	90.0	114.3	17.1	API5L-B	0/0	1.00	1.00	1	1.49	None
								1.00	1.00	2	1.49	None
J05	Elbow	152L	90.0	114.3	17.1	API5L-B	0/0	1.00	1.00	1	1.49	None
								1.00	1.00	2	1.49	None
J06	Elbow	152L	90.0	114.3	17.1	API5L-B	0/0	1.00	1.00	1	1.49	None
								1.00	1.00	2	1.49	None
K02	Elbow	229L	90.0	168.3	11.0	API5L-B	0/0	1.64	1.37	1	4.07	None
								1.64	1.37	2	4.07	None
L01	Elbow	152L	90.0	114.3	17.1	API5L-B	0/0	1.00	1.00	1	1.49	None
								1.00	1.00	2	1.49	None
L05	Elbow	152L	90.0	114.3	11.1	API5L-B	0/0	1.22	1.01	1	2.59	None
								1.22	1.01	2	2.59	None
L06	Elbow	152L	90.0	114.3	11.1	API5L-B	0/0	1.22	1.01	1	2.59	None
								1.22	1.01	2	2.59	None
M01	Elbow	229L	90.0	168.3	11.0	API5L-B	0/0	1.64	1.37	1	4.07	None
								1.64	1.37	2	4.07	None
M02	Elbow	229L	38.7	168.3	11.0	API5L-B	0/0	1.64	1.37	1	4.07	None
								1.64	1.37	2	4.07	None
N03	Elbow	76L	90.0	60.3	11.1	API5L-B	0/0	1.00	1.00	1	1.19	None
								1.00	1.00	2	1.19	None
N05	Elbow	76L	90.0	60.3	11.1	API5L-B	0/0	1.00	1.00	1	1.19	None
								1.00	1.00	2	1.19	None
N11	Elbow	76L	90.0	60.3	11.1	API5L-B	0/0	1.00	1.00	1	1.19	None
								1.00	1.00	2	1.19	None

^ = Number of bend ends where either a flange or valve is within distance
(L/D) x Nominal diameter, where L/D is defined under Tools > Model options > Edit
L = Long radius

T E E D A T A L I S T I N G

Point Name	Point Seg-Type	Pipe OD (mm)	Pipe Thk. (mm)	Material	Tee Type	**Crotch Radius (mm)	*Crotch Thick (mm)	SIF in	SIF out
D07	D-Header H-Branch	323.9 114.3	28.6 17.1	API5L-B API5L-B	Welding			1.20	1.27
E03	E-Header D-Branch	914.4 323.9	44.4 28.6	API5L-X60 API5L-B	Welding			1.70	1.94
J09	J-Header L-Branch	114.3 114.3	17.1 17.1	API5L-B API5L-B	Welding			1.00	1.00
K10	K-Header N-Branch	168.3 168.3	11.0 11.0	API5L-B API5L-B	Welding			1.43	1.57
K07	M-Header K-Branch	168.3 168.3	11.0 11.0	API5L-B API5L-B	Welding			1.43	1.57
K03	K-Header L-Branch	168.3 114.3	11.0 17.1	API5L-B API5L-B	Welding			1.43	1.57

* = Pad thickness for reinforced tee.

**= Fillet radius for Raised tee or discontinuity dist. for Thickened tee.

V A L V E D A T A L I S T I N G

Point Name	Type	Dnom (mm)	Length (mm)	Valve Weight (kg)	Auto	Symbol	Actuator Weight (kg)	DX (mm)	DY (mm)	DZ (mm)	SAF	Joint Type	SIF	Size /Avg. (mm)	Type /Max. (mm)
Tag No. : MV107 D01	CATALOG	300	964.95	1507	Yes	No					0.00	WN	1.00		
Tag No. : MV1062 D08	CATALOG	300	964.95	1507	Yes	No					0.00	WN	1.00		
Tag No. : <None> E01	CATALOG	900	2286.00	4838	Yes	No					0.00	WN	1.00		
Tag No. : MV1070 H01	CATALOG	100	456.95	145	Yes	No					2.00	WN	1.00		
Tag No. : MV1072 J01	CATALOG	100	456.95	145	Yes	No					2.00	WN	1.00		
Tag No. : MV1072 J07	CATALOG	100	500.00	145	Yes	No					2.00	WN	1.00		
Tag No. : MV1067 K04	CATALOG	150	457.20	153	Yes	No					2.00	WN	1.00		
Tag No. : MV1065 L02	CATALOG	100	456.95	145	Yes	No					2.00	WN	1.00		
Tag No. : E4R L03	CATALOG	100	457.20	181	Yes	No					5.40	WN	1.00		
Tag No. : MV1067 N01	CATALOG	50	368.05	48	Yes	No					2.00	WN	1.00		
Tag No. : D2R N06	CATALOG	50	368.30	95	Yes	No					4.80	WN	1.00		
Tag No. : D2R FB, N08	CATALOG	50	368.05	48	Yes	No					2.00	WN	1.00		

C A T A L O G V A L V E D A T A L I S T I N G

Point Name : D01
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 900

Point Name : D08
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 900

Point Name : E01
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 900

Point Name : H01
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 900

Point Name : J01
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 900

Point Name : J07
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 900

Point Name : K04
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 150

C A T A L O G V A L V E D A T A L I S T I N G

Point Name : L02
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 900

Point Name : L03
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Globe Valve
Type : Globe Valve
Main End : FL
Rating : 900

Point Name : N01
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 900

Point Name : N06
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Gate Valves
Type : Gate Valves
Main End : FL
Rating : 900

Point Name : N08
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 900

F L A N G E D A T A L I S T I N G

Point	FLG		Dnom	Rating	Flange	Bolt	Joint		Size	Type	ANSI
Name	No.	Type	(mm)		Weight	Weight	Type	SIF	/Avg.	/Max.	Check
					(kg)	(kg)			(mm)	(mm)	
Tag No. :		<None>									
D01	1	WELDNECK	300	900	168.74	84.37	WN	1.00			No
Tag No. :		<None>									
D02	1	WELDNECK	300	900	168.74	84.37	WN	1.00			No
Tag No. :		<None>									
D08	1	WELDNECK	300	900	168.74	0.00	WN	1.00			No
Tag No. :		<None>									
D09	1	WELDNECK	300	900	168.74	84.37	WN	1.00			No
Tag No. :		<None>									
E01	1	WELDNECK	900	900	1764.00	0.00	WN	1.00			No
Tag No. :		<None>									
E02	1	WELDNECK	900	900	1764.00	0.00	WN	1.00			No
Tag No. :		<None>									
E04	1	BLIND	900	900	1200.00	600.00	USER	1.00			No
Tag No. :		<None>									
H01	1	WELDNECK	100	900	23.13	0.00	WN	1.00			No
Tag No. :		<None>									
H02	1	WELDNECK	100	900	23.13	0.00	WN	1.00			No
Tag No. :		<None>									
J01	1	WELDNECK	100	900	23.13	0.00	WN	1.00			No
Tag No. :		<None>									
J02	1	WELDNECK	100	900	23.13	0.00	WN	1.00			No
Tag No. :		<None>									
J07	1	WELDNECK	100	900	23.13	0.00	WN	1.00			No
Tag No. :		<None>									
K09	1	WELDNECK	100	900	23.13	0.00	WN	1.00			No
Tag No. :		<None>									
K00	1	BLIND	150	150	12.70	0.00	USER	1.00			No
Tag No. :		<None>									
K04	1	WELDNECK	150	150	11.34	5.67	WN	1.00			No
Tag No. :		<None>									
K05	1	WELDNECK	150	150	11.34	5.67	WN	1.00			No
Tag No. :		<None>									
L02	1	WELDNECK	100	900	23.13	0.00	WN	1.00			No
Tag No. :		<None>									
L03	1	WELDNECK	100	900	23.13	0.00	WN	1.00			No
Tag No. :		<None>									
L03	2	WELDNECK	100	900	23.13	0.00	WN	1.00			No
Tag No. :		<None>									
L04	1	WELDNECK	100	900	23.13	0.00	WN	1.00			No
Tag No. :		<None>									
M03	1	WELDNECK	150	900	49.90	24.95	WN	1.00			No
Tag No. :		<None>									
M03	2	WELDNECK	150	900	49.90	24.95	WN	1.00			No
Tag No. :		<None>									
N01	1	WELDNECK	50	900	10.90	5.45	WN	1.00			No
Tag No. :		<None>									
N02	1	WELDNECK	50	900	10.90	5.45	WN	1.00			No
Tag No. :		<None>									
N06	1	WELDNECK	50	900	10.90	5.45	WN	1.00			No
Tag No. :		<None>									
N07	1	WELDNECK	50	900	10.90	5.45	WN	1.00			No

F L A N G E D A T A L I S T I N G

Point	FLG		Dnom		Flange	Bolt			Size	Type	
Name	No.	Type	(mm)	Rating	Weight	Weight	Joint	SIF	/Avg.	/Max.	ANSI
					(kg)	(kg)	Type		(mm)	(mm)	Check
Tag No. :											
N08	1	WELDNECK	50	900	10.90	5.45	WN	1.00			No
Tag No. :											
N09	1	WELDNECK	50	900	10.90	5.45	WN	1.00			No

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R E D U C E R D A T A L I S T I N G

From Point	To Point	Length (mm)	OD From (mm)	OD To (mm)	Thk From (mm)	Thk To (mm)	From Material	Cone Angle (deg)	SIF
K06	K08	105.00	168.3	114.3	11.0	17.1	API5L-B		1.00
N10	N12	270.00	60.3	168.3	11.1	11.0	API5L-B		1.00

C E N T E R O F G R A V I T Y R E P O R T

Description	Weight (kg)	X CoG (mm)	Y CoG (mm)	Z CoG (mm)
-----	-----	-----	-----	-----
Valve :	8956.79	35307.93	7929.35	-2755.91
Support :	0.00	0.00	0.00	0.00
Flange :	6781.93	32938.09	7722.60	-4088.56
Flexible Joint :	0.00	0.00	0.00	0.00
Additional Weight :	0.00	0.00	0.00	0.00
Pipes (Run, Bend, Tee, Reducer) :	11946.63	34654.24	8156.59	-2086.90
PipeTot (Valve+Support+Flange+Flex. Jt.+Add. Wt.+Pipes):	27685.35	34445.33	7976.76	-2793.68
Insulation :	0.00	0.00	0.00	0.00
Cladding :	0.00	0.00	0.00	0.00
Lining :	0.00	0.00	0.00	0.00
Contents :	4170.88	33104.40	7867.39	-2748.72
Beam :	0.00	0.00	0.00	0.00
PipeTot + Ins. :	27685.35	34445.33	7976.76	-2793.68
PipeTot + Ins. + Clad. :	27685.35	34445.33	7976.76	-2793.68
PipeTot + Ins. + Clad. + Lin. :	27685.35	34445.33	7976.76	-2793.68
PipeTot + Ins. + Clad. + Lin. + Cont. :	31856.22	34269.75	7962.44	-2787.79
PipeTot + Beam :	27685.35	34445.33	7976.76	-2793.68
PipeTot + Ins. + Beam :	27685.35	34445.33	7976.76	-2793.68
PipeTot + Ins. + Clad. + Beam :	27685.35	34445.33	7976.76	-2793.68
PipeTot + Ins. + Clad. + Lin. + Beam :	27685.35	34445.33	7976.76	-2793.68
PipeTot + Ins. + Clad. + Lin. + Cont. + Beam :	31856.22	34269.75	7962.44	-2787.79

NOTE: Fluid Density Factor is assumed as 1.0.

NOTE: COG report only includes the currently visible beams and segments.

All pipe segments are included.

No beams present in the model.

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BENTLEY
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L O A D C A S E		D E S C R I P T I O N
Load Case	Load Case Name	Description
GR	Gravity	
T1	Thermal 1	
T2	Thermal 2	
T3	Thermal 3	
W1	Wind 1	
W2	Wind 2	
W3	Wind 3	
W4	Wind 4	
P1	Press 1	
P2	Press 2	
P3	Press 3	

HYDROTEST LOAD SUMMARY

Specific Gravity of Fluid : 1.000

Hydrotest Pressure Case : P3

Hydrotest Pressure Factor : 1.500

Hydrotest Temperature Case : T3

Hydrotest Temperature Factor : 1.000

Cut-short Case : None

Include Insulation & Cladding: No

Point Summary

Pipe	Number of Points in the System			Mass	Pipe	Weight of System(kg)	
	Frame	Soil	Total			Contents	Total
163	0	0	163	0	27685.4	4170.9	31856.2

ANCHOR DATA LISTING

Point Name	Translational Stiffness			Rotational Stiffness			Release Hanger		
	KTX N/mm	KTY N/mm	KTZ N/mm	KRX N.m/deg	KRY N.m/deg	KRZ N.m/deg	X XX	Y YY	Z ZZ
B04	RIGID	RIGID	RIGID	Connected to: Ground		RIGID			
O12	RIGID	RIGID	RIGID	Connected to: Ground		RIGID			
B01	RIGID	RIGID	RIGID	Connected to: Ground		RIGID			
B05	RIGID	RIGID	RIGID	Connected to: Ground		RIGID			
J10	RIGID	RIGID	RIGID	Connected to: Ground		RIGID			
M03	RIGID	RIGID	RIGID	Connected to: Ground		RIGID			
O11	RIGID	RIGID	RIGID	Connected to: Ground		RIGID			

Point Connectivity

From	To	Comp.	Pipe Id	Length mm	DX	Offset (mm) DY	DZ	Global (mm) X	Y	Z
*** Segment C										
Origin	B04	Point	12"C06	0.00	44851.20	7500.00	0.00	44851.20	7500.00	0.00
B04	C01	Run	12"C06	58.33	0.00	700.00	0.00	44851.20	8200.00	0.00
*** Segment D										
Origin	C01	Point	12"C06	0.00	44851.20	8200.00	0.00	44851.20	8200.00	0.00
C01	D01	Junction	12"C06	1.67	0.00	20.00	0.00	44851.20	8220.00	0.00
D01	D02	Valve	12"C06	80.41	0.00	964.95	0.00	44851.20	9184.95	0.00
D02	D03	Bend	12"C06	47.77	0.00	573.25	0.00	44851.20	9758.20	0.00
D03	D04	Run	12"C06	350.00	0.00	0.00	-4200.00	44851.20	9758.20	-4200.00
D04	D05	Bend	12"C06	116.67	0.00	0.00	-1400.00	44851.20	9758.20	-5600.00
D05	D06	Bend	12"C06	188.18	0.00	-2258.20	0.00	44851.20	7500.00	-5600.00
D06	D11	Run	12"C06	112.50	-1350.00	0.00	0.00	43501.20	7500.00	-5600.00
D11	D07	Tee	12"C06	322.50	-3870.00	0.00	0.00	39631.20	7500.00	-5600.00
D07	D08	Run	12"C06	144.17	-1730.00	0.00	0.00	37901.20	7500.00	-5600.00
D08	D09	Valve	12"C06	80.41	-964.95	0.00	0.00	36936.25	7500.00	-5600.00
D09	D12	Run	12"C06	108.33	-1300.00	0.00	0.00	35636.25	7500.00	-5600.00
D12	E03	Tee	12"C06	273.99	-3287.85	0.00	0.00	32348.40	7500.00	-5600.00
*** Segment E										
Origin	O12	Point	36"C06	0.00	32348.40	7500.00	0.00	32348.40	7500.00	0.00
O12	E01	Run	36"C06	233.33	0.00	0.00	-2800.00	32348.40	7500.00	-2800.00
E01	E02	Valve	36"C06	190.50	0.00	0.00	-2286.00	32348.40	7500.00	-5086.00
E02	E03	Tee	36"C06	42.83	0.00	0.00	-514.00	32348.40	7500.00	-5600.00
E03	E04	Run	36"C06	155.00	0.00	0.00	-1860.00	32348.40	7500.00	-7460.00
*** Segment G										
Origin	B01	Point	4"C06	0.00	39631.20	7500.00	0.00	39631.20	7500.00	0.00
B01	G01	Run	4"C06	58.33	0.00	700.00	0.00	39631.20	8200.00	0.00

Point Connectivity

From	To	Comp.	Pipe Id	Length mm	Offset (mm)			Global (mm)		
					DX	DY	DZ	X	Y	Z
*** Segment H										
Origin	G01	Point	4"C06	0.00	39631.20	8200.00	0.00	39631.20	8200.00	0.00
G01	H01	Junction	4"C06	1.67	0.00	20.00	0.00	39631.20	8220.00	0.00
H01	H02	Valve	4"C06	38.08	0.00	456.95	0.00	39631.20	8676.95	0.00
H02	H03	Bend	4"C06	25.00	0.00	300.00	0.00	39631.20	8976.95	0.00
H03	H06	Run	4"C06	233.33	0.00	0.00	-2800.00	39631.20	8976.95	-2800.00
H06	H04	Bend	4"C06	233.33	0.00	0.00	-2800.00	39631.20	8976.95	-5600.00
H04	D07	Tee	4"C06	123.08	0.00	-1476.95	0.00	39631.20	7500.00	-5600.00
*** Segment I										
Origin	B05	Point	4"C06	0.00	47751.20	7500.00	0.00	47751.20	7500.00	0.00
B05	I01	Run	4"C06	58.33	0.00	700.00	0.00	47751.20	8200.00	0.00
*** Segment J										
Origin	I01	Point	4"C06	0.00	47751.20	8200.00	0.00	47751.20	8200.00	0.00
I01	J01	Junction	4"C06	1.67	0.00	20.00	0.00	47751.20	8220.00	0.00
J01	J02	Valve	4"C06	38.08	0.00	456.95	0.00	47751.20	8676.95	0.00
J02	J03	Bend	4"C06	25.00	0.00	300.00	0.00	47751.20	8976.95	0.00
J03	J16	Run	4"C06	250.00	0.00	0.00	3000.00	47751.20	8976.95	3000.00
J16	J04	Bend	4"C06	91.67	0.00	0.00	1100.00	47751.20	8976.95	4100.00
J04	J11	Run	4"C06	290.00	-3480.00	0.00	0.00	44271.20	8976.95	4100.00
J11	J12	Run	4"C06	290.00	-3480.00	0.00	0.00	40791.20	8976.95	4100.00
J12	J13	Run	4"C06	290.00	-3480.00	0.00	0.00	37311.20	8976.95	4100.00
J13	J14	Run	4"C06	290.00	-3480.00	0.00	0.00	33831.21	8976.95	4100.00
J14	J15	Run	4"C06	183.33	-2200.00	0.00	0.00	31631.21	8976.95	4100.00
J15	J05	Bend	4"C06	106.67	-1280.01	0.00	0.00	30351.20	8976.95	4100.00
J05	J06	Bend	4"C06	54.17	0.00	650.00	0.00	30351.20	9626.95	4100.00
J06	J09	Tee	4"C06	50.00	-600.00	0.00	0.00	29751.20	9626.95	4100.00

Point Connectivity

From	To	Comp.	Pipe Id	Length mm	DX	Offset (mm) DY	DZ	Global (mm) X Y		Z
J09	J10	Run	4"C06	25.00	-300.00	0.00	0.00	29451.20	9626.95	4100.00
J10	J07	Run	4"C06	25.00	-300.00	0.00	0.00	29151.20	9626.95	4100.00
J07	K09	Valve	4"C06	41.67	-300.00	400.00	0.00	28851.20	10026.95	4100.00
*** Segment K										
Origin	K00	Point	6"B06	0.00	26056.09	9651.95	2557.00	26056.09	9651.95	2557.00
K00	K10	Tee	6"B06	20.83	0.00	0.00	250.00	26056.09	9651.95	2807.00
K10	K11	Run	6"B06	53.87	0.00	0.00	646.50	26056.09	9651.95	3453.50
K11	K07	Tee	6"B06	53.88	0.00	0.00	646.50	26056.09	9651.95	4100.00
K07	K02	Bend	6"B06	31.25	0.00	375.00	0.00	26056.09	10026.95	4100.00
K02	K03	Tee	6"B06	29.17	350.00	0.00	0.00	26406.09	10026.95	4100.00
K03	K12	Run	6"B06	45.83	550.00	0.00	0.00	26956.09	10026.95	4100.00
K12	K04	Run	6"B06	45.83	550.00	0.00	0.00	27506.09	10026.95	4100.00
K04	K05	Valve	6"B06	38.10	457.20	0.00	0.00	27963.29	10026.95	4100.00
K05	K06	Run	6"B06	62.74	752.91	0.00	0.00	28716.20	10026.95	4100.00
K06	K08	Reducer	4"C06	8.75	105.00	0.00	0.00	28821.20	10026.95	4100.00
K08	K09	Run	4"C06	2.50	30.00	0.00	0.00	28851.20	10026.95	4100.00
*** Segment L										
Origin	J09	Tee	4"C06	0.00	29751.20	9626.95	4100.00	29751.20	9626.95	4100.00
J09	L01	Bend	4"C06	33.33	0.00	0.00	-400.00	29751.20	9626.95	3700.00
L01	L08	Run	4"C06	56.25	-675.00	0.00	0.00	29076.20	9626.95	3700.00
L08	L02	Run	4"C06	56.25	-675.00	0.00	0.00	28401.20	9626.95	3700.00
L02	L03	Valve	4"C06	38.08	-456.95	0.00	0.00	27944.25	9626.95	3700.00
L03	L04	Valve	4"B06	38.10	-457.20	0.00	0.00	27487.05	9626.95	3700.00
L04	L09	Run	4"B06	45.04	-540.48	0.00	0.00	26946.57	9626.95	3700.00
L09	L05	Bend	4"B06	45.04	-540.49	0.00	0.00	26406.08	9626.95	3700.00
L05	L06	Bend	4"B06	33.33	0.00	400.00	0.00	26406.08	10026.95	3700.00

Point Connectivity

From	To	Comp.	Pipe Id	Length mm	DX	Offset (mm) DY	DZ	Global (mm) X	Y	Z
L06	K03	Tee	4"C06	33.33	0.00	0.00	400.00	26406.09	10026.95	4100.00
*** Segment M										
Origin	K07	Tee	6"B06	0.00	26056.09	9651.95	4100.00	26056.09	9651.95	4100.00
K07	M09	Run	6"B06	58.33	0.00	0.00	700.00	26056.09	9651.95	4800.00
M09	M08	Run	6"B06	162.50	0.00	0.00	1950.00	26056.09	9651.95	6750.00
M08	M01	Bend	6"B06	41.67	0.00	0.00	500.00	26056.09	9651.95	7250.00
M01	M02	Bend	6"B06	80.04	-750.00	-600.00	0.00	25306.09	9051.95	7250.00
M02	M07	Run	6"B06	170.83	-2050.00	0.00	0.00	23256.09	9051.95	7250.00
M07	M06	Run	6"B06	416.67	-5000.00	0.00	0.00	18256.09	9051.95	7250.00
M06	M05	Run	6"B06	416.67	-5000.00	0.00	0.00	13256.09	9051.95	7250.00
M05	M04	Run	6"B06	416.67	-5000.00	0.00	0.00	8256.09	9051.95	7250.00
M04	M03	Run	6"B06	416.67	-5000.00	0.00	0.00	3256.09	9051.95	7250.00
*** Segment N										
Origin	O11	Point	2"C06	0.00	29848.40	7500.00	0.00	29848.40	7500.00	0.00
O11	N04	Run	2"C06	33.33	0.00	400.00	0.00	29848.40	7900.00	0.00
N04	N01	Run	2"C06	8.33	0.00	100.00	0.00	29848.40	8000.00	0.00
N01	N02	Valve	2"C06	30.67	0.00	368.05	0.00	29848.40	8368.05	0.00
N02	N03	Bend	2"C06	37.50	0.00	450.00	0.00	29848.40	8818.04	0.00
N03	N16	Run	2"C06	166.67	0.00	0.00	2000.00	29848.40	8818.04	2000.00
N16	N05	Bend	2"C06	67.25	0.00	0.00	807.00	29848.40	8818.04	2807.00
N05	N11	Bend	2"C06	69.49	0.00	833.90	0.00	29848.40	9651.95	2807.00
N11	N06	Run	2"C06	29.17	-350.00	0.00	0.00	29498.40	9651.95	2807.00
N06	N07	Valve	2"C06	30.69	-368.30	0.00	0.00	29130.10	9651.95	2807.00
N07	N15	Run	2"C06	39.58	-475.00	0.00	0.00	28655.10	9651.95	2807.00
N15	N08	Run	2"C06	39.58	-475.00	0.00	0.00	28180.10	9651.95	2807.00
N08	N09	Valve	2"B06	30.67	-368.05	0.00	0.00	27812.05	9651.95	2807.00

Point Connectivity

From	To	Comp.	Pipe Id	Length mm	Offset (mm)			Global (mm)		
					DX	DY	DZ	X	Y	Z
N09	N10	Run	2"B06	33.33	-400.00	0.00	0.00	27412.05	9651.95	2807.00
N10	N12	Reducer	6"B06	22.50	-270.00	0.00	0.00	27142.05	9651.95	2807.00
N12	N14	Run	6"B06	43.17	-517.98	0.00	0.00	26624.07	9651.95	2807.00
N14	K10	Tee	6"B06	47.33	-567.98	0.00	0.00	26056.09	9651.95	2807.00

